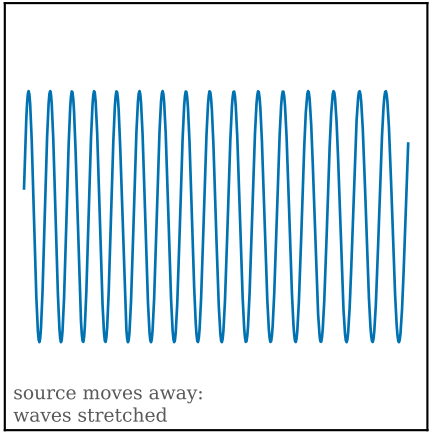
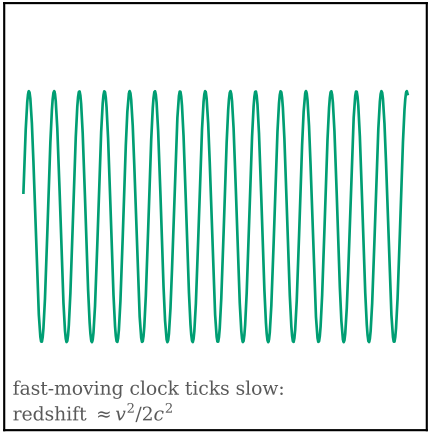
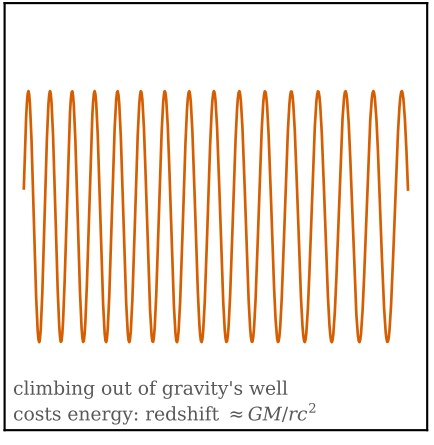

classical Doppler
source moves away:
waves stretched


transverse Doppler
fast-moving clock ticks slow:
redshift $\approx v^2/2c^2$


gravitational
climbing out of gravity's well
costs energy: redshift $\approx GM/rc^2$